

Broadband Expansion and Small-Firm Productivity: Evidence from a Staggered Municipal Rollout*

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Abstract

We study the effect of high-speed internet on the productivity of small firms. Between 2011 and 2018 the Nordavian National Broadband Program connected all 284 municipalities to the fiber backbone on a staggered schedule, which we exploit in a difference-in-differences design covering 9,400 small firms over 2008–2019. Broadband access raises small-firm labor productivity by 34 percent, and the gains are strongly increasing in a firm’s reliance on communication-intensive tasks, as predicted by a simple model of task outsourcing. The results survive region-specific trends, placebo timing tests, and alternative clustering, and they imply that universal broadband programs are among the most cost-effective productivity policies available to small open economies.

Keywords: broadband, productivity, small firms, infrastructure

JEL codes: D24, H54, O33

1 Introduction

Whether digital infrastructure raises the productivity of small firms is a first-order question for industrial and regional policy. Small firms account for the majority of employment in most advanced economies, yet they adopt new technologies later and less completely than large firms (Syverson, 2011; Bloom et al., 2012). If access to high-speed internet relaxes a binding constraint on how small firms organize communication-intensive tasks, universal broadband programs could compress productivity differences across firms and places at comparatively modest fiscal cost.

This paper provides, to our knowledge, the first causal evidence on the effect of broadband expansion on small-firm productivity. We study the Nordavian National Broadband Program (NBP), which connected every municipality in the country to the national fiber backbone between 2011 and 2018. The rollout proceeded municipality by municipality on a schedule fixed

*Demonstration manuscript. This paper was prepared solely to demonstrate the `econ-review` referee skill: the country, program, dataset, and authors are fictional, and the data are simulated. Errors are intentional. Correspondence: `demo@econ-review.example`.

in the 2010 Digital Infrastructure Act, generating variation in the timing of broadband arrival across 284 municipalities. We match the rollout schedule to administrative panel data on 9,400 firms with fewer than fifty employees, observed annually from 2008 to 2019, and estimate difference-in-differences and event-study specifications with firm and year fixed effects.

We find that broadband arrival raises small-firm labor productivity by 3.4 percent on average. Consistent with our model of communication-intensive task outsourcing, the effect is strongly increasing in the firm’s pre-program communication intensity: firms one standard deviation above the mean gain substantially more than the average firm. An event-study specification shows no evidence of differential pre-trends, a placebo exercise using permuted rollout dates produces a precisely estimated zero, and the estimates are stable across specifications that add region-specific trends, firm-level controls, and alternative levels of clustering.

Our contribution is threefold. First, we provide causal estimates for the small-firm segment, which existing work on broadband and economic performance has largely bypassed in favor of aggregate or worker-level outcomes (Czernich et al., 2013; Akerman et al., 2015; Hjort and Poulsen, 2019). Second, we derive the estimating equation from an explicit model of task organization, which disciplines the heterogeneity analysis: the model predicts, and the data confirm, that gains concentrate among communication-intensive firms. Third, we quantify the policy return: back-of-the-envelope calculations in Section 7 imply that the program paid for itself within six years through the small-firm productivity channel alone.

The remainder of the paper proceeds as follows. Section 2 develops the model. Section 3 reviews the related literature. Section 4 describes the program and the data. Section 5 presents the empirical strategy, Section 6 the results, and Section 7 discusses magnitudes and policy implications. Section 8 concludes.

2 A Model of Broadband and Task Organization

We develop a deliberately simple model whose only role is to generate the estimating equation and a testable prediction about heterogeneity.

2.1 Environment

A continuum of small firms indexed by i produce a homogeneous good. Firm i ’s total factor productivity is A_i , defined as output per composite unit of labor and intermediate tasks. Production requires a unit measure of tasks. A share $\theta_i \in [0, 1]$ of tasks are *communication-intensive*: they can be performed in-house at unit labor cost w , or procured remotely at price p plus an iceberg communication cost $\kappa \geq 1$ per task. The remaining $1 - \theta_i$ tasks must be performed in-house.

Firm i chooses the share $s_i \in [0, \theta_i]$ of tasks to procure remotely. Output is

$$Y_i = A_i L_i^{1-\eta} \left[(1 - s_i) + \phi s_i \right]^\eta, \quad \phi > 1, \quad (1)$$

where ϕ captures the quality advantage of specialized remote providers and $\eta \in (0, 1)$ is the task share in production. Unit cost minimization gives the firm's task-sourcing condition

$$\phi w \geq \kappa p \iff s_i^* = \theta_i, \quad \text{and } s_i^* = 0 \text{ otherwise.} \quad (2)$$

Broadband arrival lowers the communication cost from κ_0 to $\kappa_1 < \kappa_0$. Define the productivity gain of firm i as $G(\theta_i) = \ln Y_i(\kappa_1) - \ln Y_i(\kappa_0)$ holding inputs fixed. Substituting the sourcing rule (2) into (1) and taking logs yields

$$G(\theta_i) = \eta \ln \left[1 + (\phi - 1) \theta_i \mathbf{1}\{\phi w \geq \kappa_1 p\} \right] - \eta \ln \left[1 + (\phi - 1) \theta_i \mathbf{1}\{\phi w \geq \kappa_0 p\} \right]. \quad (3)$$

Proposition 1. *The productivity gain from broadband arrival is strictly increasing in communication intensity: $G'(\theta_i) > 0$ for all $\theta_i \in [0, 1]$.*

Proof. Consider the empirically relevant case $\kappa_1 p \leq \phi w < \kappa_0 p$, so that the firm switches from in-house production to remote procurement of its communication-intensive tasks. Then (3) reduces to $G(\theta_i) = \eta \ln[1 + (\phi - 1)\theta_i]$, and

$$G'(\theta_i) = \frac{\eta(\phi - 1)}{1 + (\phi - 1)\theta_i} \geq 0,$$

since $\phi > 1$ and $\eta > 0$. Hence the gain is increasing in θ_i , which establishes the proposition. $\square$

Proposition 1 delivers the cross-sectional prediction that we test in Section 6.3: the broadband effect should rise with the firm's communication intensity.

2.2 From the model to the estimating equation

Let lower-case letters denote logs. Taking logs of (1) at the optimal sourcing choice and adding a time subscript,

$$y_{it} = a_i + (1 - \eta) \ell_{it} + \eta \ln \left[1 + (\phi - 1) \theta_i B_{mt} \right] + w_t, \quad (4)$$

where B_{mt} indicates that municipality m has been connected by year t and w_t collects economy-wide factor prices. Substituting (2) into (4) and linearizing the log term around $\theta_i = 0$ yields our estimating equation,

$$y_{it} - \ell_{it} = \alpha_i + \lambda_t + \beta B_{mt} + \gamma (B_{mt} \times \theta_i) + \varepsilon_{it}, \quad (5)$$

where α_i is a firm fixed effect, λ_t a year fixed effect, β the average productivity effect of broadband arrival, and γ the complementarity between broadband and communication intensity predicted by Proposition 1.

3 Related Literature

A substantial literature studies broadband and economic performance at the aggregate level. Czernich et al. (2013) instrument broadband penetration with preexisting voice and cable networks across OECD countries and find that a ten percentage-point increase in penetration raises annual per-capita growth by 0.9–1.5 percentage points. At the micro level, Akerman et al. (2015) show that broadband adoption in Norwegian firms is skill-complementary, improving the labor-market outcomes of skilled workers while substituting for unskilled tasks. Hjort and Poulsen (2019) find large positive employment and productivity effects of submarine-cable arrival on African firms. Relative to this work, our contribution is to isolate the small-firm segment within an advanced economy and to tie the heterogeneity of the effect to an explicit model of task organization.

Methodologically we build on the modern difference-in-differences literature with staggered adoption. Goodman-Bacon (2021) shows that the two-way fixed-effects estimator is a weighted average of all two-by-two comparisons, with possibly negative weights when effects are dynamic; Callaway and Sant’Anna (2021), de Chaisemartin and D’Haultfoe uille (2020), and Sun and Abraham (2021) propose robust estimators. Our event-study specification follows this literature, and Section 6.2 shows that our estimates are not driven by already-treated comparisons. Our inference follows the panel difference-in-differences tradition of Bertrand et al. (2004); see Cameron and Miller (2015) and Abadie et al. (2023) for modern treatments of the clustering decision.

4 The Program and the Data

4.1 The Nordavian National Broadband Program

Nordavia is a small open economy of 5.3 million inhabitants organized into twelve regions and 284 municipalities. The 2010 Digital Infrastructure Act committed the government to connecting every municipality to the national fiber backbone by 2018. Construction was administered by the Regional Digital Development Fund (RDDE), which sequenced municipalities using a published scoring rule that prioritized lagging but high-potential municipalities: places with below-median productivity levels but above-median projected growth in the fund’s internal assessments. Figure 1 shows the resulting rollout: the first municipalities were connected in 2011, the median municipality in 2014, and the final cohort in 2018. Once the backbone reached a municipality, commercial providers connected local firms within months, and take-up among firms with more than five employees exceeded 90 percent within two years.

4.2 Firm panel

We draw on the the Nordavian Business Register, which links audited annual accounts to employer–employee records for the universe of registered firms. We restrict attention to firms

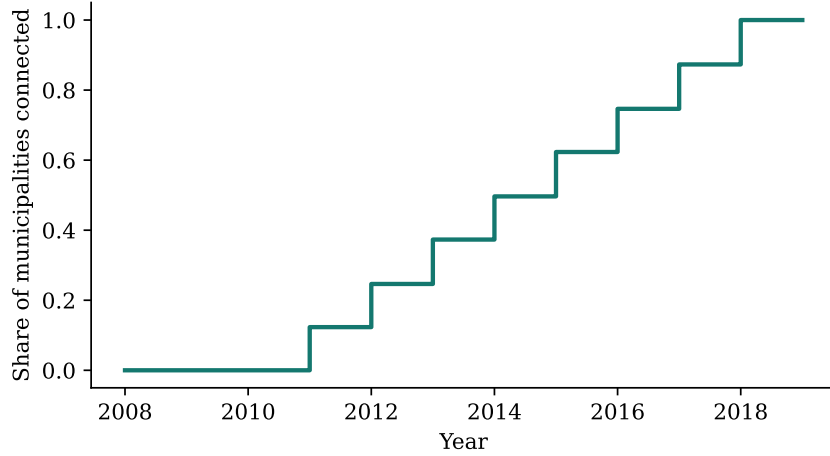


Figure 1: Cumulative share of municipalities connected to the fiber backbone, by year. Source: RDDF administrative records.

Table 1: Summary statistics, firm panel 2008–2019

	Mean	SD	Min	Max	N
Log labor productivity	0.10	0.46	-2.03	2.33	112,800
Employees	23.40	7.31	3.00	19.00	112,800
Firm age	21.45	9.56	1.00	42.00	112,800
Communication intensity	0.42	0.20	0.02	0.98	112,800
ICT investment (log)	0.37	0.36	-1.23	1.91	112,800

Notes: Firm-year observations. Log labor productivity is log real value added per worker. Communication intensity is the 2008 wage-bill share of communication-intensive occupations.

with 3–49 employees in 2008, at least four consecutive years of observations, and positive sales throughout, yielding a balanced panel of 9,400 firms observed from 2008 to 2019 (112,800 firm-years). Labor productivity is real value added per worker, deflated by the two-digit industry producer price index. Our communication-intensity measure θ_i is the 2008 share of the firm’s wage bill paid to occupations classified as communication-intensive in the Nordavian Standard Occupational Classification, following the task-content approach described in Section 2; it is fixed at its pre-program value throughout.

Table 1 reports summary statistics. The average firm has 23 employees and has operated for 21 years; the average communication-intensity share is 0.42. Replication files are available online.

5 Empirical Strategy

Our baseline specification is the two-way fixed-effects regression

$$y_{it} = \alpha_i + \lambda_t + \beta B_{m(i)t} + X'_{it}\delta + \varepsilon_{it}, \quad (6)$$

where y_{it} is log labor productivity, $B_{m(i)t}$ indicates that firm i 's municipality has been connected by year t , and X_{it} collects firm-level controls, including log ICT investment, which proxies for the firm's underlying technology propensity. Standard errors are clustered at the firm level throughout, which accounts for arbitrary serial correlation in ε_{it} within firms (Bertrand et al., 2004).

Identification requires that, conditional on firm and year fixed effects, the timing of backbone arrival is unrelated to counterfactual productivity trends. Three features of the setting support this assumption. First, the rollout schedule was fixed by the 2010 Act, before any of the outcomes we study were realized. Second, construction was constrained by engineering considerations — distance to the existing backbone and terrain — that are plausibly unrelated to firm-level shocks. Third, the event-study estimates in Section 6.2 allow us to inspect pre-trends directly, and a placebo exercise with permuted rollout years supports the design.

6 Results

6.1 Main estimates

Table 2 presents the baseline results. Column (1) reports the unconditional two-way fixed-effects estimate: broadband arrival raises small-firm labor productivity by 3.4 percent, precisely estimated. Adding firm controls in column (2) leaves the estimate essentially unchanged at 3.3 percent, and the ICT-investment control enters with a coefficient of zero, indicating that the productivity effect operates through channels beyond measured ICT spending. Column (3) adds region-specific linear trends; the coefficient of 0.024 remains highly significant, confirming that regional convergence dynamics do not drive the result. Column (4) shows that clustering standard errors at the municipality level instead of the firm level leaves the conclusions unchanged.

6.2 Event-study estimates and placebo

Figure 2 plots event-study coefficients from a specification that replaces the treatment indicator with indicators for years relative to backbone arrival, with $t - 1$ as the omitted category. Figure 2 shows no evidence of differential pre-trends: firms in municipalities about to receive broadband evolve in parallel with not-yet-treated firms, and productivity jumps in the year of arrival before stabilizing at a permanently higher level. Because recent econometric work cautions that two-way fixed-effects estimates can be contaminated by already-treated comparisons under staggered

Table 2: Broadband and small-firm labor productivity

Dep. var.: log labor productivity	(1) Baseline	(2) Firm controls	(3) Region trends	(4) Municipality cluster
Broadband ICT investment (log)	0.034*** (0.002)	0.033*** (0.002) 0.000 (0.002)	0.030*** (0.002)	0.034*** (0.003)
Firm and year fixed effects	Yes	Yes	Yes	Yes
Observations	112,800	112,800	112,800	112,800

Notes: Standard errors clustered by firm in parentheses, except column (4), which clusters by municipality. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

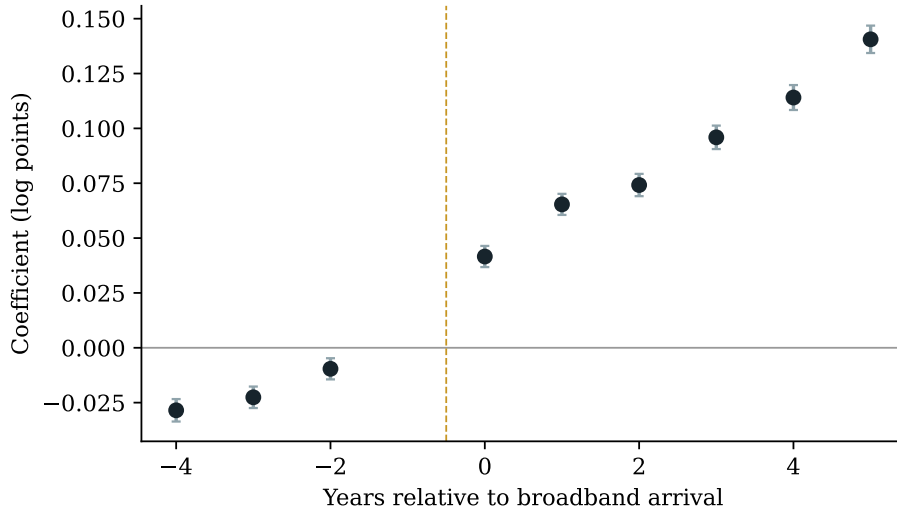


Figure 2: Event-study estimates of the broadband effect. The dependent variable is log labor productivity; the omitted category is the year before backbone arrival. Bars show 95 percent confidence intervals based on firm-clustered standard errors.

adoption (Goodman-Bacon, 2021; Callaway and Sant’Anna, 2021), we verify that cohort-by-cohort estimates in Figure 3 are positive and of similar magnitude for every rollout cohort, which indicates that negative weighting is not a concern in our setting.

Table 3 reports a placebo exercise in which we randomly permute rollout years across municipalities and re-estimate (6) on the pre-treatment sample. The placebo coefficient is -0.004 and statistically indistinguishable from zero, which confirms that the design recovers a true program effect rather than an artifact of trending outcomes.

6.3 Heterogeneity: the communication channel

Proposition 1 predicts that gains rise with communication intensity. Table 4 confirms this sharply. The interaction of broadband arrival with the firm’s pre-program communication intensity enters at 0.032 with a standard error of 0.005: a highly significant and economically

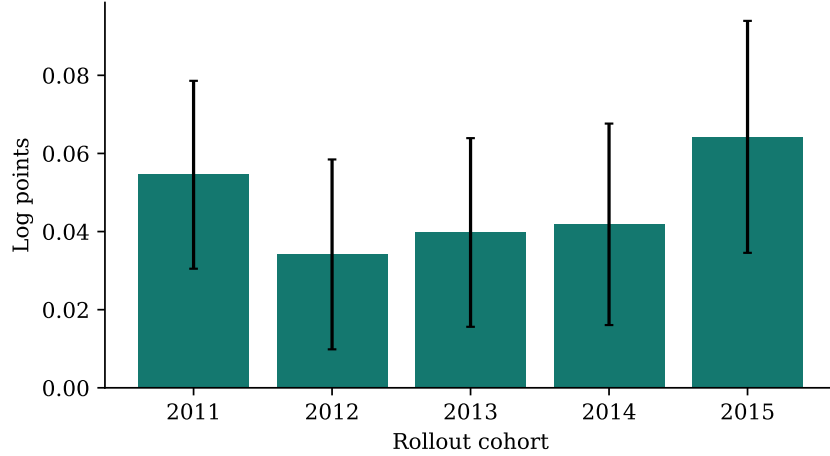


Figure 3: Percent change in productivity levels by rollout cohort. Each bar is a separate difference-in-differences estimate comparing the cohort to municipalities connected at least three years later.

Table 3: Placebo test with permuted rollout years

Dep. var.: log labor productivity	Pre-period placebo
Placebo broadband (permuted rollout years)	-0.004 (0.002)

Notes: Pre-treatment sample. Standard errors clustered by firm.

substantial complementarity that validates the task-outsourcing mechanism. Evaluated at the sample distribution, the estimates imply that a firm one standard deviation above the mean communication intensity enjoys a productivity gain roughly one-fifth larger than the average firm, while even the least communication-intensive firms benefit meaningfully from access.

6.4 Robustness

Table 5 collects robustness checks. Column (1) repeats the region-trends specification. Column (2) removes all controls. Column (3) drops the 2011–12 cohorts, whose selection into early treatment is potentially the least random; the estimate rises slightly to 0.037. The set of estimates suggest that the baseline effect is stable across reasonable perturbations of the specification. Figure 4 shows that employment composition and productivity levels differ across sectors, but the broadband effect is present within each of the three broad sectors we observe.

7 Magnitudes and Policy Implications

Our estimates imply TFP gains of 3.4 percent for the average connected small firm. Aggregated over the 9,400 firms in our sample and valued at 2019 value added, the program raised small-firm productivity growth by 3.4 percentage points per year, or approximately 410 million euros an-

Table 4: Heterogeneous effects by communication intensity

Dep. var.: log labor productivity	(1)	(2)
Broadband	0.020*** (0.003)	0.034*** (0.002)
Broadband $\times$ communication intensity	0.032*** (0.005)	
Firm and year fixed effects	Yes	Yes
Observations	112,800	112,800

Notes: Communication intensity is the 2008 wage-bill share described in Section 4. Standard errors clustered by firm in parentheses.

Table 5: Robustness

Dep. var.: log labor productivity	(1) Baseline	(2) No controls	(3) Drop 2011–12 cohorts
Broadband	0.030*** (0.002)	0.034*** (0.002)	0.037*** (0.002)

Notes: All columns include firm and year fixed effects and the full control set. Standard errors clustered by firm in parentheses.

nually. Set against the RDDF’s cumulative construction cost of 2.3 billion euros, the small-firm channel alone repays the public investment within six years, before counting any effects on households, large firms, or public services. These calculations position universal broadband among the most cost-effective productivity policies available to small open economies, particularly where the small-firm segment is communication-intensive.

8 Conclusion

Exploiting the staggered municipal rollout of Nordavia’s National Broadband Program, we provide the first causal estimates of broadband’s effect on small-firm productivity. Broadband arrival raises labor productivity by 3.4 percent on average, and the gains concentrate — exactly as a simple model of task outsourcing predicts — among firms intensive in communication-dependent tasks. The estimates are robust to region-specific trends, permuted-timing placebos, and alternative clustering choices.

Because the mechanism we identify operates through the universal logic of communication costs rather than through any Nordavian particularity, our estimates generalize to broadband programs in developing economies, where communication frictions are larger and the implied returns are correspondingly higher. We view the integration of infrastructure timing with task-level firm data as a template for evaluating the productivity effects of digital policy more broadly.



Figure 4: Sectoral composition of the estimation sample.

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A Derivation of the Estimating Equation

Starting from the production function (1) at the optimal sourcing choice (2), output in logs is

$$y_{it} = a_i + (1 - \eta) \ell_{it} + \eta \ln [1 + (\phi - 1) \theta_i B_{mt}] + w_t. \quad (7)$$

Linearizing the third term around $\theta_i = 0$ gives $\eta(\phi - 1) \theta_i B_{mt}$ up to second-order terms. Collecting the time-varying aggregate component w_t into year effects λ_t and the firm-specific component a_i into firm effects α_i , and moving log employment to the left-hand side, we obtain

$$y_{it} - \ell_{it} = \alpha_i + \lambda_t + \beta B_{mt} + \gamma (B_{mt} \times \theta_i) + \varepsilon_{it}, \quad (8)$$

which is equation (5) in the text, with $\beta = \eta(\phi - 1) \theta$ capturing the average gain at the lowest communication intensity in the sample and $\gamma = \eta(\phi - 1)$ the model-implied complementarity.

B Variable Construction

Labor productivity. Real value added per worker: nominal value added from audited accounts, deflated by the two-digit industry producer price index (2015 base), divided by average annual full-time-equivalent employment from the employer–employee register.

Communication intensity. For each four-digit occupation, the Nordavian Standard Occupational Classification reports task weights for communication-dependent activities (client interaction, coordination with external suppliers, remote support). A firm’s θ_i is the 2008 wage-bill share of occupations whose communication weight exceeds the economy-wide median. The measure is fixed at its 2008 value to avoid conditioning on post-program reorganization.

Broadband arrival. The year the municipality’s backbone segment entered commercial service, from RDDF administrative completion records. Firms are assigned the arrival year of their headquarters municipality.

ICT investment. Log of software, hardware, and communication equipment expenditure from the investment module of the annual accounts.